Extended Reality in Nuclear Engineering: A Scoping Review Protocol

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INTRODUCTION

Nuclear power plants and facilities involve complex, safety-critical operations that demand high situational awareness, procedural compliance, and decision-making accuracy. Personnel in plant operations, maintenance, safeguards, radiation work, emergency response, and training must continuously interact with equipment, monitoring systems, procedures, and large volumes of technical information [1]. As nuclear facilities adopt advanced digital technologies, improving human-system interaction has become an important research area for operational efficiency, safety, and workforce performance.

Extended Reality (XR), encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), has emerged as a promising technology for human-centered activities in complex engineering environments, supporting procedural guidance, equipment maintenance, remote collaboration, digital twin integration, radiation visualization, and operator training [2]. Prior studies across industrial domains have demonstrated XR's potential to improve operational performance, knowledge transfer, and human factors outcomes. A recent bibliometric review of 700 articles on AI and VR in high-risk training simulations reported a 5.54% average annual increase in publications between 2015 and 2025, identifying nuclear as one of four major thematic clusters alongside emergency medicine, aviation, and disaster response [8].

XR also continues to gain commercial momentum independent of any single platform: global shipments of XR hardware, including headsets and smart glasses, grew 41.6% in 2025 to 14.5 million units, with double-digit annual growth forecast through 2030 as AI-enabled smart glasses and spatial computing expand the technology's reach into industrial settings [11]. This reinforces XR's relevance as a maturing, task-specific tool for training, maintenance, and operations support in domains such as nuclear engineering.

Within nuclear engineering, research has explored XR for reactor operations, maintenance, safeguards and verification, radiation work, emergency preparedness, and workforce training. Mixed Reality has been investigated for safeguards inspections through remote expert assistance and guided inspection procedures, and XR integrated with digital twins has shown promise for monitoring, predictive maintenance, and human performance in complex nuclear systems [3]. Industrial deployment studies beyond the nuclear sector further underscore the importance of systematic validation prior to operational implementation.

Despite this growing interest, existing research remains fragmented across operations, safeguards, radiation work, and training, reported using diverse technologies, methods, and objectives. Existing reviews have begun to address this fragmentation but remain limited in scope or currency. A 2016 review organized VR applications in nuclear research into eight domains, including operator training, dose assessment, physical security, and reactor design, but was restricted to VR alone, emphasized Brazilian research, and covered literature only through 2015, predating modern AR/MR platforms and XR-integrated digital twins [9]. A 2024 scoping review synthesized VR training solutions for nuclear power plant field operators, but likewise did not extend to AR or MR, safeguards, radiation visualization, or emergency response [6]. There is currently no comprehensive synthesis mapping XR applications across the full breadth of nuclear engineering activity or identifying research gaps to guide future deployment. This protocol describes a scoping review designed to address that gap.

The objective of this scoping review is to systematically examine the current landscape of XR applications within nuclear engineering and to identify emerging trends, implementation strategies, evaluation approaches, and future research opportunities across operations, safeguards, radiation work, and training. The review is guided by four research questions: what technological approaches and system design strategies have emerged for implementing XR (RQ1); what technical, operational, and human-factor challenges influence the development and deployment of XR systems (RQ2); what are the effects of using XR compared with conventional methods on operational performance, learning outcomes, and human factors (RQ3); and what research gaps and future research opportunities should guide the next generation of XR research in this domain (RQ4).

METHODS

This scoping review will be conducted according to the Joanna Briggs Institute (JBI) methodology for scoping reviews [4], which comprises six stages: (1) identifying the research question, (2) identifying relevant studies, (3) study selection, (4) charting the data, (5) collating, summarizing,

and reporting the results, and (6) stakeholder consultation. The review will be reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) checklist [5]. The protocol will be registered on the Open Science Framework prior to the start of screening. The review is planned to begin in fall 2026 and conclude within approximately six months.

Stage 1: Identifying the Research Question

The objective of this review is to outline the technological approaches, design strategies, challenges, and evaluation outcomes reported for XR applications in nuclear engineering, and to identify gaps to guide future research. In line with this objective, the review addresses four research questions: (1) What technological approaches and system design strategies have emerged for implementing XR in nuclear engineering? (2) What technical, operational, and human-factor challenges influence the development and deployment of XR systems in this domain? (3) What are the effects of using XR compared with conventional methods on operational performance, learning outcomes, and human factors? (4) What research gaps and future research opportunities should guide the next generation of XR research in nuclear engineering?

The population of interest includes any study describing the design, implementation, or evaluation of an XR system used by nuclear engineering personnel, including operators, maintenance technicians, safeguards inspectors, radiation protection staff, emergency responders, and trainees. The concept of interest is the application of VR, AR, or MR technology to operational, safeguards, radiation-related, emergency response, or workforce training tasks within the nuclear sector. The context is broadly defined and will include studies conducted in all geographical locations, all facility types (e.g., commercial light-water reactors, research reactors, advanced reactors and small modular reactors, fuel cycle facilities), and all study designs, including experimental, observational, qualitative, and case-study work.

Stage 2: Identifying Relevant Studies

A comprehensive search will be conducted across six electronic databases selected to span engineering, computing, and human factors literature: IEEE Xplore, Scopus, Web of Science, the ACM Digital Library, INSPEC, and the American Nuclear Society digital library. Grey literature will be searched through ProQuest Dissertations & Theses, conference proceedings (e.g., ANS Winter and Annual Meetings, IEEE VR, ISMAR), and a Google Scholar search of the top 100 results.

The search strategy will combine two concept groups using the Boolean operator AND, with terms within each group combined using OR. The first group captures XR terminology: “virtual reality,” “augmented reality,” “mixed reality,” “extended reality,” “head-mounted display,” and named platforms (e.g., “HoloLens,” “Oculus,” “Vive”). The second group captures nuclear-domain terminology: “nuclear power plant,” “reactor operations,” “safeguards,” “nuclear inspection,” “radiation protection,” “emergency response,” “operator training,” and “nuclear workforce.” The strategy will be piloted in IEEE Xplore and refined for precision before being adapted to the syntax of each remaining database, with the assistance of a research librarian. Backward and forward reference (snowball) searching will be performed on all included studies to identify additional sources. Searches will be limited to English-language publications, with no lower date restriction given the relative novelty of consumer XR hardware. All records will be imported into EndNote, and duplicates will be removed prior to screening.

Stage 3: Study Selection

Study selection will follow a two-step screening process: title and abstract screening, followed by full-text review. Two reviewers (D.K. and H.K.) will independently screen all records against the eligibility criteria in Table I. Prior to full screening, a calibration exercise will be conducted on a random 5% sample of records to confirm adequate inter-rater agreement (Cohen’s kappa ≥ 0.6); the criteria will be revised if agreement falls below this threshold. Disagreements at either stage will be resolved through discussion, with a third reviewer (J.S.) arbitrating unresolved conflicts. Reasons for exclusion at the full-text stage will be recorded and reported in a PRISMA flow diagram.

Stage 4: Charting the Data

Data will be extracted independently by D.K. and H.K. using a structured form developed in Microsoft Excel and piloted on a sample of 3–5 included studies prior to full extraction. Extracted fields will include: (a) study characteristics – author, year, country, facility type, study design, sample size, and participant role (e.g., operator, technician, inspector, trainee); (b) technology characteristics – XR modality (VR/AR/MR), hardware platform, display type, tracking method, level of immersion, interactivity, and integration with digital twins or other systems; and (c) application and outcome characteristics – operational domain (operations, safeguards, radiation work, emergency response, training), target task, evaluation method, comparator (if any), and reported outcomes related to performance, learning, or human factors. Disagreements in extraction will be resolved by discussion, with J.S. acting as arbiter where consensus cannot be reached. Study authors will be contacted directly for missing or unclear data where feasible.

Stage 5: Collating, Summarizing, and Reporting Results

Extracted data will be summarized descriptively using frequency counts and cross-tabulations (e.g., XR modality by operational domain, evaluation method by publication year) to characterize the breadth of the field. Results will be presented in tabular and diagrammatic form, including a

bubble or heat-map style visualization mapping XR modality against the four operational domains (operations, safeguards, radiation work, and training). A narrative synthesis will accompany the data map, organized around the four research questions, to characterize technological and system design trends (RQ1), recurring technical, operational, and human-factor challenges (RQ2), reported comparative effects on performance and learning outcomes (RQ3), and gaps warranting future research (RQ4). Where studies report quantitative outcome measures, effect directions will be tabulated descriptively; given the heterogeneity expected across study designs, no meta-analysis or formal quality appraisal of evidence is planned, consistent with standard scoping review practice.

Parameter	Inclusion	Exclusion
Technology	XR (VR/AR/MR) is the primary technology studied	XR is incidental or not the primary technology
Application	Application is within nuclear engineering (operations, safeguards, radiation, emergency response, training)	Application falls outside the nuclear sector
Publication type	Peer-reviewed journal or conference paper	Opinion piece, editorial, or abstract without methodological detail
Language	English language; full text available	Non-English; full text unavailable

TABLE I: Inclusion and Exclusion Criteria

Stage 6: Consultation

Consistent with the consultation exercise recommended by Levac et al. as a core component of scoping review methodology, preliminary findings will be presented to a panel of nuclear industry and human factors subject-matter experts, including operations and training personnel from the project's industry partner. This consultation is intended to validate the completeness of the technology and application mapping, contextualize findings against current industry practice, and refine the identified research gaps prior to final dissemination. Feedback from this exercise will be incorporated into the final manuscript and acknowledged separately from the formal review findings.

RESULTS

This protocol serves as the methodological foundation for a forthcoming scoping review. By prospectively detailing our search, selection, and data charting strategy a priori, we aim to ensure the rigor, transparency, and reproducibility of our synthesis of the XR literature in nuclear engineering. To the best of the authors' knowledge, this will be the first scoping review to map XR applications across the full breadth of nuclear engineering activity, rather than focusing on a single domain such as training or maintenance in isolation. The completed review is expected to yield a comprehensive map of XR applications across operations, safeguards, radiation protection, and training, organized by technology type, system design strategy, and evaluation approach.

This mapping is anticipated to surface methodological inconsistencies in how XR systems are evaluated for operational performance and learning transfer, particularly the extent to which studies validate transfer from simulated XR environments to real-world operational settings. This concern is not hypothetical: in a related review of 135 immersive VR training studies across industrial domains, only 29% found VR superior to comparator training methods, 21% reported positive prospects without demonstrated superiority, and 36% reported some learning potential with no comparison group at all [7]. A nuclear-specific VR review reached a similar conclusion, finding that the benefit of VR training over conventional methods was highly conditional on fidelity, learner guidance, and alignment with training goals, rather than guaranteed by the technology itself [6]. The review is also expected to clarify where individualized, adaptive XR-based training complements rather than substitutes for team-level operational performance, an important framing distinction given that nuclear operations are inherently team-based. Furthermore, the review is expected to identify which nuclear application domains have received disproportionate research attention relative to others, providing a basis for prioritizing future research investment and deployment. This pattern has historical precedent: an earlier review of VR applications across eight nuclear domains found that operator training/ergonomics and dose assessment together accounted for half of all reviewed publications, while domains such as reactor design, reactor refueling, and waste treatment were comparatively underrepresented despite their continued relevance [9].

The review is also expected to surface regulatory asymmetries that shape XR adoption across roles. Prior work has noted that main control room operator training is tightly bound to validated full-scope physical simulators, such that XR can supplement but not yet substitute for this training, whereas training for field-based and maintenance personnel is comparatively less regulated and may therefore adopt XR more readily [6]. Mapping where such regulatory constraints differ across operational domains is expected to clarify which parts of the nuclear workforce represent the most near-term feasible targets for XR adoption. A related challenge is anticipated, as a broader VR review found nuclear platforms under-disclosed relative to other sectors, often citing non-

disclosure agreements, alongside an absence of haptic feedback [10].

The research gaps identified by this review (RQ4) are also likely to echo broader patterns observed across high-risk training research generally, even outside the nuclear sector specifically. A bibliometric review spanning 700 articles on AI and VR in high-risk training across multiple industries found that only 18% addressed the ethical aspects of biometric or sensitive data collected during training, and that 73% of studies focused narrowly on sectoral applications without engaging unified theoretical frameworks [8]. While these figures describe the high-risk training literature broadly rather than nuclear engineering specifically, they suggest that data-ethics considerations and theoretical integration may be similarly underdeveloped within the nuclear-specific literature this review will examine, a possibility the data extraction process is designed to evaluate directly.

A primary strength of this protocol is its scope across four domains, identifying cross-domain trends that single-domain reviews would miss; a limitation is that formal quality appraisal will not be performed, so the review characterizes the breadth of evidence rather than its reliability. By consolidating fragmented findings into a single framework, this review aims to help researchers, training developers, and regulators identify where XR offers clear value and where further validation is required before deployment.

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